Prince Gajnani

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Education

Indian Institute of Information Technology Gwalior **Aug 2022 – Jun 2027**
Dual degree : B.Tech + M.tech in Information Technology *CGPA - 8.12*

Experience

Software Engineer Intern **May 2025 – Jul 2025**
B.P Technologies *Remote*

- Engineered a **distributed task scheduler** in **Node.js** that sustained **5,000+ jobs/minute** by integrating **Redis Streams** with consumer groups for fault-tolerant, high-volume background job processing.
- Optimized the core **transaction-logging API** by restructuring the **PostgreSQL** schema with **covering and partial indexes** and **CTEs**, significantly reducing **query latency** on historical data retrieval.
- Built a **real-time monitoring dashboard** in **React.js** that renders **10,000+ active log entries** without blocking the UI thread by implementing **chunked data streaming** and **DOM virtualization**.

Projects

Matching Engine Analyser | *Rust, Tokio, Actix-Web, Docker, Kubernetes, Redis, TCP*

- Engineered a **concurrent, high-throughput trading-engine benchmarking platform** in **Rust (Tokio/Actix-Web)** that auto-builds and grades submissions inside **isolated Docker containers** via a semaphore-bounded job queue.
- Architected a **3-phase grading pipeline** (correctness, throughput, burst) stress-testing a **100K+ order** async bot fleet over **TCP** to benchmark **p50/p90/p99 tail latency** and throughput into a composite score.
- Delivered a **real-time leaderboard** using **Server-Sent Events** and **Redis** persistence, containerized with **Docker Compose & Kubernetes** for reproducible, scalable deployment.

ClubSphere | *React.js, Node.js, Express.js, MongoDB, Socket.io, JWT, AWS Bedrock, Cloudinary, Tailwind CSS*

- Built a full-stack **MERN club management platform** with **10+ RESTful API modules** (clubs, events, posts, messaging, notifications) featuring a role-based **join-request approval workflow**.
- Implemented **JWT authentication**, real-time **Socket.io messaging and notifications**, **Cloudinary** media uploads, and automated event reminders via **cron jobs** to drive member engagement.
- Integrated **AI-powered features** using **AWS Bedrock (Claude Haiku)** for intelligent chatbot support and personalized event recommendations, and optimized the backend with **Redis caching**, **database indexing**, **pagination**, and **rate limiting** for scalable, low-latency performance.

Technical Skills

Languages: C++, Javascript, Rust, C, SQL
Frameworks & Libraries: React.js, Node.js, Express.js, Actix-Web, Tailwind CSS
Databases: MySQL, PostgreSQL, MongoDB, Redis, Mongoose
Cloud & DevOps: Docker, Kubernetes, AWS (S3, Bedrock), Git, GitHub, Postman
Core CS: Data Structures & Algorithms, OOP, REST APIs, WebSockets, Operating Systems, DBMS, Computer Networks

Achievements

- Solved **1900+** Data Structures and Algorithms and Competitive Programming problems across multiple platforms.
- Secured a rank of **398** in the **ICPC 2025** Preliminary Rounds.
- Secured a rank of **870** in the **IICPC Codefest 2026** Preliminary Round among top intercollegiate participants.
- Highest rating on Codeforces - **1500+**, Specialist [Profile] | **4-Star** on CodeChef [Profile]
- **LeetCode Knight** badge with a rating of **2000+** (Top **2.53%**) [Profile]
- Selected as a **Top 50 Finalist (out of 400,000 participants) in the Google Big Code 2026 Hackathon.**
- Qualified **GATE CSE 2026** with an AIR of **2472** and **GATE CSE 2025** with an AIR of **6858**.

Position of Responsibility

Vice Chairman, IEEE Students Branch **2024 – 2025**
IIITM Gwalior *Gwalior, India*

- Led the **IEEE Students Branch** core team as **Vice Chairman** for one year, planning and executing technical workshops, coding events, and community initiatives to boost student engagement.